

# Week 1 Lab Task: Enhance the Counter App

**Duration:** 90 minutes | Individual submission | Submit via GitHub before the deadline announced in class

## Learning Objectives

By completing this task, you will be able to:

- ☐ Modify an existing StatefulWidget's UI and internal state
- ☐ Add a second interactive button and wire it to setState()
- ☐ Conditionally render a widget based on state
- ☐ Personalise ThemeData using colorSchemeSeed
- ☐ Push a working Flutter project to a public GitHub repository

## Before You Start

This task builds directly on the default flutter create counter app. You are not building from scratch — you are editing a known-good pattern (see Lecture Slides 15–16). Read every task below carefully: two of the values you need are personal to you and are not in any generic tutorial, ChatGPT answer, or classmate's repository. Copying someone else's code without adapting these values will produce visibly wrong output and will be flagged during the milestone review (per Course Policy 8.3 — Academic Integrity & AI Tools).

## Your Personal Parameters

Calculate these once, at the top of your main.dart, as a comment block:

| Parameter   | How to Calculate It                                                                                                                                                                            |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| myThreshold | Sum of the last 3 digits of your university roll number, plus 5                                                                                                                                |
| mySeedColor | The Colors. value whose first letter matches the first letter of your first name (e.g., Ayesha → Colors.amber; if no colour starts with your letter, use the letter of your last name instead) |

*You will use both values in the tasks below. Graders will check your GitHub commit against your roll number and name — mismatched values are treated as an unverified or incomplete submission.*

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**For Initializing the Project if you have not already done yet.**

```
# Step 1: Create a new project
$ flutter create my_first_app
# Step 2: Open in VS Code
$ cd my_first_app
$ code .
# Step 3: Run the app
$ flutter run
# Or press F5 in VS Code
```

.....

## Tasks

### 1. Add a Reset Button

Add a second Floating Action Button that resets the counter to 0 when tapped. Use Icons.refresh. Arrange both Floating Action Buttons so neither overlaps the other.

### 2. Personalised Threshold Message

Below the counter number, display “You're on a roll!” in green — but only when the count exceeds your myThreshold value (not a hardcoded 10). Hide the message otherwise.

### 3. Reset Counter Tracker

Add a new state variable that tracks how many times the Reset Floating Action Button has been pressed. Display it on screen as: Resets used: N. This requires adding a second piece of state and updating it inside a second setState() call — don't reuse the increment logic.

### 4. Personalised Theme

In ThemeData, set colorSchemeSeed: mySeedColor (from your Personal Parameters table above, not Colors.indigo from the lecture example).

### 5. About Line

Add a small Text widget near the bottom of the screen showing: Built by <Your Full Name> · <Roll Number>.

### 6. Written Reflection (in your README, not the code)

In 2–4 sentences, in your own words, explain what setState() {...} does and why the app doesn't update the screen without it. One-line or copy-pasted definitions will not receive credit — this is checked for understanding, not just presence.

## Submission Requirements

- ☐ Create a new public GitHub repository (not private, not a fork of a classmate's repository).
- ☐ Your repository's README.md must include your name and roll number, a screenshot of the running app showing the counter above your personal threshold (message visible) and at least one reset performed, and your written reflection from Task 6.
- ☐ Push all code — confirm lib/main.dart reflects all six tasks.
- ☐ Submit the repository link to the instructor before the deadline announced in class. Late submissions follow the penalty in Course Policy 8.2 (20% per day).
- ☐ Do not commit unrelated files (build artifacts) — .gitignore should exclude /build, .dart\_tool, etc.

## Time Budget (Target: 90 Minutes)

| Task                                  | Estimated Time |
|---------------------------------------|----------------|
| Calculate personal parameters         | 3 min          |
| Reset Floating Action Button (Task 1) | 15 min         |
| Threshold message (Task 2)            | 15 min         |
| Reset tracker (Task 3)                | 20 min         |
| Theme + About line (Tasks 4–5)        | 10 min         |
| Test on emulator / simulator          | 10 min         |
| Write reflection + README + push      | 15 min         |
| Buffer                                | 2 min          |

## Grading Checklist

Pass/Fail per item — contributes to Weekly Lab Tasks (20% of final grade):

- ☐ Two Floating Action Buttons present, non-overlapping, correct icons
- ☐ Threshold message uses myThreshold, not a hardcoded number
- ☐ Reset counter tracked and displayed correctly
- ☐ colorSchemeSeed matches mySeedColor
- ☐ About line present with correct name / roll number
- ☐ README contains screenshot + reflection
- ☐ Repository is public and link submitted before deadline
- ☐ Reflection shows genuine understanding (not a copy-pasted definition)

**Submit to:** your instructor, via the link-submission method announced in class — before the deadline announced in class.